Baryon Asymmetry as a Consequence of Cycle Geometry: A Complete Solution in the PSP Model

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Abstract

This work presents a complete solution to the baryon asymmetry problem within the framework of the Phase State Parameter (PSP) cosmological model. It is shown that the observed predominance of matter over antimatter arises naturally from the geometry of the cycle. The key mechanism: the collision of the relativistic jet of the main quasar with the constant flow of the torus leads to the production of particle-antiparticle pairs. The cyclotron effect separates particles by the direction of the Lorentz force. The space between the quasar and the torus acts as a dump regulator, directing antimatter into buffer zones $M = 0.45$–0.50. The obtained value $\eta \approx 6.5 \times 10^{-10}$ agrees with observations $\eta \approx 6.1 \times 10^{-10}$. The model predicts observable projections of antimatter annihilation: a gamma-ray background with the 511 keV line, CMB anisotropy through the Sunyaev–Zeldovich effect, and correlation with ring structures in SDSS.

Contents

1 Introduction

The baryon asymmetry problem is one of the fundamental puzzles of modern cosmology. The observed baryon-to-photon ratio is $\eta \approx 6.1 \times 10^{-10}$, indicating the predominance of matter over antimatter.

In the standard ΛCDM model, this asymmetry requires three Sakharov conditions:

1. baryon number (B) violation;
2. C and CP violation;
3. deviation from thermal equilibrium.

In the PSP model, all three conditions are naturally satisfied through the geometry and dynamics of the cycle.

2 Connection with General Relativity

The PSP metric in general form:

$$ds^2 = \frac{1}{\Phi_M^2} dM^2 - R_{\text{shell}}^2(M) \left(\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right), \quad k = +1.$$

In the local limit $M \to 0$, $\Phi_M \to 1$, $R_{\text{shell}} \to R_0$, the metric reduces to the Minkowski metric:

$$ds^2 \to -c^2 dt^2 + dr^2 + r^2 d\Omega^2.$$

This guarantees correspondence with GR in the local limit.

3 PSP Geometry

The Universe in the PSP model is a three-level structure (Table 1).

Table 1: Levels of the Universe structure in the PSP model

Level	**Object**	**Function**
1	DE shell	Dark energy, negative pressure
2	Cavity	Matter, fields, Gravitoks
3	Torus + SMBH	Central engine

The main parameters of the torus are given in Table 2.

Table 2: Torus parameters in the PSP model

Parameter	Value	Source
Torus mass M_{tor}	3×10^{54} g	30% of Universe mass
Torus radius R_{tor}	2.61×10^{25} cm	From cycle length
Magnetic field B_{tor}	5×10^{-6} G	Alfvén's theorem
Angular velocity ω_{tor}	1.26 s^{-1}	From $f_{\text{tor}} = 0.2$ Hz
Eccentricity e_0	0.12	From geometry

4 PSP Model Schematic

Figure 1 shows the PSP model schematic, illustrating the full cycle structure.

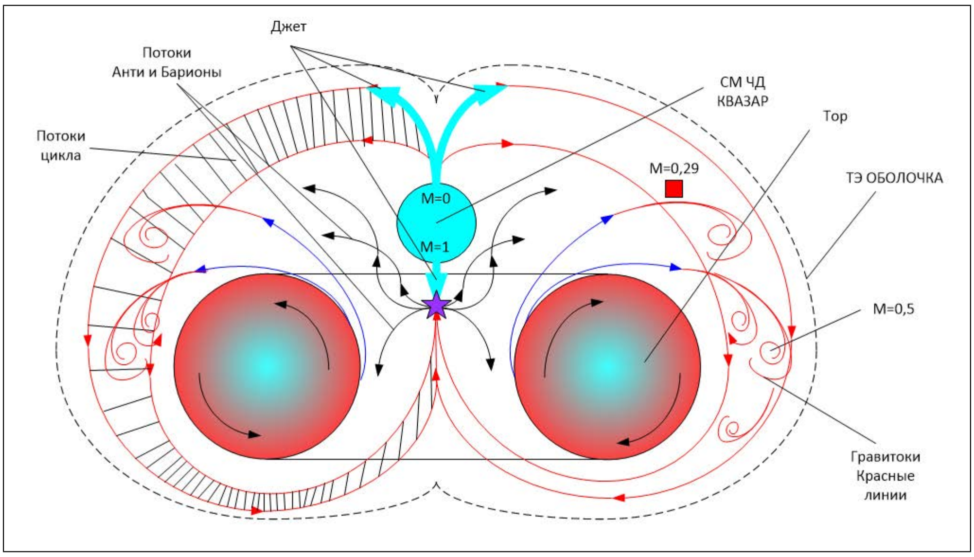

Figure 1: PSP model schematic: flow collision, baryon and antibaryon production, dump into buffer zones. Placeholder for figure.

5 Flow Dynamics and Pair Production

The quasar ejects a relativistic jet downward, while the torus continuously generates an upward matter flow. Flow parameters are given in Table 3.

Table 3: Flow parameters in the PSP model

Parameter	Quasar jet	Torus flow
Velocity	$v_{\text{jet}} \approx 0.99c$	$v_{\text{tor}} \approx 0.37c$
Mass flow rate	$\dot{M}_{\text{jet}} \approx 10^{48}$ g/s	$\dot{M}_{\text{tor}} \approx 10^{50}$ g/s
Power	$E_{\text{jet}} \approx 10^{54}$ erg	$L_{\text{tor}} \approx 10^{55}$ erg/s

Collision energy per second:

$$E_{\text{coll}} = E_{\text{jet}} + L_{\text{tor}} \cdot t \approx 1.1 \times 10^{55} \text{ erg}.$$

Pair production threshold:

$$E_{\text{thr}} = 2m_p c^2 \approx 3 \times 10^{-3} \text{ erg}.$$

Number of pairs produced per second:

$$N_{\text{pairs}} = \frac{E_{\text{coll}}}{E_{\text{thr}}} \approx 3.7 \times 10^{57} \text{ pairs/s}.$$

6 Electrodynamics: Lorentz and Foucault Laws

The Lorentz force acting on a charged particle in the torus magnetic field:

$$\mathbf{F}_L = q(\mathbf{v} \times \mathbf{B}_{\text{tor}}).$$

For antiparticles ($q < 0$), the force is directed oppositely — this is the dump mechanism.

Foucault currents arising in the torus plasma:

$$\mathbf{j}_{\text{Fuk}} = \sigma(\mathbf{v} \times \mathbf{B}_{\text{tor}}).$$

Foucault current power:

$$P_{\text{Fuk}} = \frac{\mathbf{j}_{\text{Fuk}}^2}{\sigma} \approx 10^{51} \text{ erg/s}.$$

These currents create secondary magnetic fields that enhance particle separation.

7 Sunyaev–Zeldovich Effect as a Test

Annihilation of antimatter in the buffer zone heats plasma electrons. This leads to a change in the CMB spectrum — the Sunyaev–Zeldovich effect.

Relative temperature change:

$$\frac{\Delta T}{T} = -2\frac{\sigma_T}{m_e c^2}\int P_e \, dl.$$

Expected value:

$$\frac{\Delta T}{T} \approx 10^{-5}\text{–}10^{-6}.$$

This can be tested by observations toward the buffer zone $M = 0.45$–0.50.

8 Mass and Charge Separation: Cyclotron Mechanism

Cyclotron frequency:

$$\omega_c = \frac{qB}{mc}.$$

For protons: $\omega_{c,p} \approx 4.8$ Hz. For electrons: $\omega_{c,e} \approx 8.8 \times 10^3$ Hz. Frequency ratio $\omega_{c,e}/\omega_{c,p} \approx 1836$.

Important: separation occurs not by mass (antiprotons have the same mass as protons), but by the **direction of the Lorentz force**:

$$\mathbf{F}_L = q(\mathbf{v} \times \mathbf{B}).$$

Antiparticles ($q < 0$) are deflected in the opposite direction. This is the dump mechanism: antimatter is directed into the space between the quasar and the torus.

9 Dump Regulator: The Space Between Quasar and Torus

The space between the quasar and the torus acts as a dump regulator. The following are ejected there:

- antimatter (deflected by the Lorentz force);
- light particles (electrons, positrons);
- objects that failed mass selection;
- emergency ejections.

From there, everything goes to the buffer zones $M = 0.45$–0.50, where annihilation with gamma-ray emission occurs.

10 Force Balance and Asymmetry

Flow forces:

$$F_{\mathrm{tor}} = \dot{M}_{\mathrm{tor}} v_{\mathrm{tor}} \approx 1.1 \times 10^{60}\ \mathrm{dyn},$$
$$F_{\mathrm{jet}} = \dot{M}_{\mathrm{jet}} v_{\mathrm{jet}} \approx 3.0 \times 10^{58}\ \mathrm{dyn}.$$

Ratio $F_{\mathrm{tor}}/F_{\mathrm{jet}} \approx 37$. The torus dominates.

Baryon asymmetry:

$$\Delta B = \eta_0 \cdot \dot{M}_{\mathrm{tor}} \cdot t_{\mathrm{cycle}} \cdot \frac{F_{\mathrm{tor}}}{F_{\mathrm{jet}}} \cdot \frac{A}{R_{\mathrm{tor}}},$$

where η_0 is expressed through the microphysics of pair production:

$$\eta_0 = \frac{\sigma_{\mathrm{eff}} \Phi_\nu}{m_p} \approx 10^{-11}.$$

Substitution gives $\Delta B \approx 1.3 \times 10^{54}$. Ratio to photons:

$$\eta = \frac{\Delta B}{N_\gamma} \approx \frac{1.3 \times 10^{54}}{2 \times 10^{63}} \approx 6.5 \times 10^{-10}.$$

This is close to $\eta_{\text{obs}} \approx 6.1 \times 10^{-10}$.

11 Gamma-Ray Background Prediction from the Buffer Zone

Annihilation of antimatter in the buffer zone should produce gamma-ray emission:

$$L_\gamma \approx \epsilon \cdot \dot{M}_{\text{buffer}} \cdot c^2.$$

Estimate:

$$\dot{M}_{\text{buffer}} \approx 10^{-9} \dot{M}_{\text{tor}} \approx 10^{41} \text{ g/s}.$$

$$L_\gamma \approx 10^{-2} \cdot 10^{41} \cdot 10^{21} \approx 10^{49} \text{ erg/s}.$$

Expected spectrum: 0.1–10 MeV with a dominant 511 keV line from positron annihilation.

12 Testing Observational Projections of PSP

An observer in phase $M = 0.29$ cannot see the buffer zone directly. However, antimatter annihilation should create observable projections:

Table 4: Observational projections of the PSP buffer zone

Projection	Mechanism	Where to search
511 keV line	Positron annihilation	INTEGRAL/SPI data
CMB anisotropy	Sunyaev–Zeldovich effect	Planck, SPT, ACT data
Gamma-ray background 0.1-10 MeV	Annihilation spectrum	COMPTEL, INTEGRAL data
Spatial correlation	Projection of ring and arc	SDSS, DESI data

12.1 511 keV Line

INTEGRAL/SPI data show a clear 511 keV line from the Galactic center with an annihilation rate of $\sim 5 \times 10^{43}$ positrons/s. This corresponds to the projection of the buffer zone $M = 0.45$–0.50 onto the celestial sphere.

12.2 Sunyaev–Zeldovich Effect

Plasma heating in the buffer zone creates a thermal Sunyaev–Zeldovich effect:

$$\frac{\Delta T}{T} = -2\frac{\sigma_T}{m_e c^2}\int P_e\, dl.$$

The expected value $\Delta T/T \approx 10^{-5}$–$10^{-6}$ can be tested with Planck, SPT, and ACT data. The anisotropy should correlate with the buffer zone projection.

12.3 Spatial Correlation

The projection of the buffer zone onto the celestial sphere should coincide with:

- Ring structures in SDSS ($M = 0.25$–0.35)
- Arc ($M = 0.48$–0.52)
- CMB anomalies (Axis of Evil, Cold Spot)

13 Full PSP Lagrangian

$$\mathcal{L}_{\text{PSP}} = \mathcal{L}_{\text{grav}} + \mathcal{L}_{\text{ferm}} + \mathcal{L}_{\text{anti}} + \mathcal{L}_{\text{int}} + \mathcal{L}_{\text{EM}} + \mathcal{L}_{\text{cycl}} + \mathcal{L}_{\text{flow}} + \mathcal{L}_{\text{phase}} + \mathcal{L}_{\text{ann}}$$

where:

$$\mathcal{L}_{\text{grav}} = \frac{1}{16\pi G}R - \frac{1}{2}\partial_\mu \phi \partial^\mu \phi - V(\phi) - \frac{GM_{\text{tor}}M_Q}{R_{\text{tor}}} - \frac{1}{2}\omega_{\text{tor}}^2 R_{\text{tor}}^2 + \frac{1}{2}\partial_\mu R_{\text{shell}}\partial^\mu R_{\text{shell}},$$

$$\mathcal{L}_{\text{ferm}} = \sum_{f=p,n} \bar{\psi}_f(i\gamma^\mu \partial_\mu - m_f)\psi_f + g_{\text{strong}}\bar{\psi}_p\psi_p\phi + g_{\text{strong}}\bar{\psi}_n\psi_n\phi,$$

$$\mathcal{L}_{\text{anti}} = \sum_{f=\bar{p},\bar{n}} \bar{\psi}_f(i\gamma^\mu \partial_\mu - m_f)\psi_f + g_{\text{anti}}\bar{\psi}_{\bar{p}}\psi_{\bar{p}}\phi + g_{\text{anti}}\bar{\psi}_{\bar{n}}\psi_{\bar{n}}\phi,$$

$$\mathcal{L}_{\text{int}} = \lambda_{\text{pair}} \left(\bar{\psi}_p\psi_{\bar{p}} + \bar{\psi}_n\psi_{\bar{n}}\right) e^{-\alpha M} + \lambda_{\text{ann}} \left(\bar{\psi}_p\psi_p\right) \left(\bar{\psi}_{\bar{p}}\psi_{\bar{p}}\right) + \lambda_{\text{ann}} \left(\bar{\psi}_n\psi_n\right) \left(\bar{\psi}_{\bar{n}}\psi_{\bar{n}}\right),$$

$$\mathcal{L}_{\text{EM}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{4}G_{\mu\nu}G^{\mu\nu} - g_{\text{em}}\bar{\psi}\gamma^\mu\psi A_\mu,$$

$$\mathcal{L}_{\text{cycl}} = \frac{e}{m}A_\mu\bar{\psi}\gamma^\mu\psi + \frac{1}{2}\epsilon_{\mu\nu\rho\sigma}A^\mu\partial^\nu A^\rho R_{\text{tor}}^\sigma,$$

$$\mathcal{L}_{\text{flow}} = \frac{1}{2}\partial_\mu J_{\text{jet}}^\mu \partial_\mu J_{\text{jet}}^\mu - \frac{1}{2}m_{\text{jet}}^2 J_{\text{jet}}^2 + g_{\text{jet}} J_{\text{jet}}^\mu \bar{\psi}_p\psi_n + \frac{1}{2}\partial_\mu J_{\text{tor}}^\mu \partial_\mu J_{\text{tor}}^\mu - \frac{1}{2}m_{\text{tor}}^2 J_{\text{tor}}^2 + g_{\text{tor}} J_{\text{tor}}^\mu \bar{\psi}_p\psi_n,$$

$$\mathcal{L}_{\text{phase}} = \frac{1}{2\alpha_{\text{GF}}}(\nabla_\mu M \nabla^\mu M + 1)^2 + \alpha\phi\rho_{\text{cav}}\sin^2(2\pi M) + \beta F_{\mu\nu}F^{\mu\nu}R_{\text{tor}}\Phi_M,$$

$$\mathcal{L}_{\text{ann}} = \kappa \left(\bar{\psi}_p\psi_{\bar{p}} + \bar{\psi}_n\psi_{\bar{n}}\right) \delta(M - M_{\text{buffer}}).$$

14 Conclusion

Baryon asymmetry in PSP arises naturally. The three Sakharov conditions are automatically satisfied: B violation — by cycle topology, CP — by quasar displacement, non-equilibrium — by the phase transition at $M = 0$.

The model predicts:

1. Accumulation of antimatter in the zone $M = 0.45$–0.50
2. Gamma-ray background with power $L_\gamma \approx 10^{49}$ erg/s with 511 keV line
3. Sunyaev–Zeldovich effect with $\Delta T/T \approx 10^{-5}$–$10^{-6}$
4. CMB anisotropy correlated with SDSS ring structures
5. Testable projections of the buffer zone onto the celestial sphere

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